This ICCV Workshop paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

QualiVision: Multi-Modal Video Quality Assessment with Quality-Aware Fusion and Discriminative Learning Strategies

Ritik Bompilwar
Khoury College of Computer Sciences
Northeastern University
bompilwar.r@northeastern.edu

Saurabh Koshatwar
Viterbi School of Engineering
University of Southern California
skoshatw@usc.edu

Abstract

The proliferation of AI-generated video content across media production, entertainment, and social platforms has created new challenges in automatic quality assessment. While traditional metrics focus primarily on technical artifacts, AI-generated content requires comprehensive evaluation across temporal consistency, image fidelity, aesthetic appeal, and text-video alignment to ensure semantic coherence between input prompts and visual outputs. We present QualiVision, a multi-modal video quality assessment framework designed to address these requirements. Working with the VQualA 2025 challenge dataset of 4,000 annotated AI-generated videos, we develop two complementary architectures that integrate video understanding with textual prompt analysis. Our first approach enhances DOVER++ with a quality-aware cross-modal fusion mechanism that dynamically adapts attention based on textual content. The second employs V-JEPA2 with strategic layer freezing and discriminative learning rates for efficient adaptation of large-scale transformers. Both models utilize a hybrid loss function combining smooth L1, ranking, and scale-aware components to better align with human perceptual judgments. Experimental evaluation of our V-JEPA2 architecture achieved a final score of 0.57, and DOVER++ obtained 0.39 on the challenge test set, demonstrating the effectiveness of cross-modal understanding for AI-generated video quality assessment.

1. Introduction

The AI-generated video market is experiencing unprecedented growth, with market valuations projected to reach USD 1.9 billion by 2030 from USD 554.9 million in 2023, representing a CAGR of 19.9% [21]. More aggressive estimates suggest the market could reach USD 2.5 billion by 2032 at a CAGR of 19.5% [1]. This explosive growth, driven by increasing demand for personalized content across media production, entertainment, and social platforms, underscores the critical need for robust quality assessment frameworks. As AI-generated videos become integral to digital marketing strategies, educational content, and creative workflows, ensuring their quality becomes paramount for user experience and commercial viability.

To support this rapidly expanding ecosystem, sophisticated evaluation methodologies are essential. Recent research has begun addressing this challenge through specialized competitions and frameworks. The NTIRE 2024 Quality Assessment of AI-Generated Content Challenge introduced systematic approaches for both image and video quality assessment of AI-generated content, with 514 registered participants across image and video tracks [15]. Similarly, the AIS 2024 Video Quality Assessment Challenge focused on User-Generated Content, demonstrating the growing recognition of quality assessment as a fundamental research area [4]. However, these efforts primarily addressed traditional quality metrics and user-generated content, leaving a gap for comprehensive evaluation of AI-generated videos with multi-modal understanding.

The unique characteristics of AI-generated video content necessitate evaluation across four critical dimensions: temporal consistency, image fidelity, aesthetic appeal, and text-video alignment. Unlike traditional video quality assessment that primarily focuses on compression artifacts and technical distortions [22], AI-generated content requires understanding the semantic relationship between input prompts and visual outputs. Temporal consistency ensures smooth motion and coherent scene transitions, while image fidelity evaluates the photorealistic quality of individual frames. Aesthetic appeal captures the artistic and visual attractiveness of the content, and text-video alignment measures how well the generated video corresponds to the input textual description. This multi-dimensional framework addresses the comprehensive quality requirements for AI-generated video applications.

Our work presents QualiVision, a multi-modal video quality assessment framework specifically designed for AI-

generated content evaluation. We develop two complementary architectures that integrate video understanding with textual prompt analysis. The first approach enhances DOVER++ (Disentangled Objective Video Quality Evaluator) [30] with a quality-aware cross-modal fusion mechanism that dynamically adapts attention based on textual content. The second employs V-JEPA2 (Video Joint Embedding Predictive Architecture) [2] with strategic layer freezing and discriminative learning strategies for efficient adaptation of large-scale transformers.

The key contributions of this work include:

1. A quality-aware cross-modal fusion mechanism that dynamically weights different quality aspects based on textual content
2. A discriminative learning strategy for large-scale video transformers with strategic layer freezing
3. A hybrid loss function combining smooth L1 loss, ranking loss, and scale-aware components for improved correlation with human perceptual judgments
4. Evaluation on the VQualA 2025 challenge set demonstrating the effectiveness of cross-modal understanding for AI-generated video quality assessment

The remainder of this paper is organized as follows: Section 2 reviews related work in video quality assessment and multimodal fusion approaches, Section 3 presents our multi-modal architectures and training strategies, Section 4 describes the experimental setup and implementation details, Section 5 presents quantitative and qualitative results, and Section 6 concludes with future directions.

2. Related Work

2.1. Traditional Video Quality Assessment

Video quality assessment has been a fundamental research area in computer vision and multimedia processing for decades. Traditional approaches can be categorized into full-reference (FR), reduced-reference (RR), and no-reference (NR) methods based on the availability of reference information [22]. Early metrics such as Peak Signal-to-Noise Ratio (PSNR) and Mean Squared Error (MSE) provided simple mathematical measures but failed to correlate well with human perceptual quality judgments.

The introduction of Structural Similarity Index (SSIM) [26] marked a significant advancement by incorporating luminance, contrast, and structural information to better align with human visual perception. SSIM evaluates the structural similarity between reference and distorted images by comparing local patterns of pixel intensities, making it more perceptually meaningful than pixel-wise error metrics.

More recent developments include the Video Multimethod Assessment Fusion (VMAF) metric [14], developed by Netflix to address the limitations of existing quality metrics for diverse video content. VMAF combines multiple elementary metrics using machine learning approaches to predict subjective video quality more accurately. The Visual Information Fidelity (VIF) criterion [23] leverages natural scene statistics and information fidelity principles to assess image quality degradation.

In addition, recent broad surveys synthesize progress across subjective protocols and objective PVQA models, spanning images, videos, immersive media, and GenAI scenarios [19, 34, 36]. For a foundational perspective focused on perceptual video quality metrics and human vision factors, see Winkler's classic review [27].

2.2. Deep Learning Approaches for Quality Assessment

The advent of deep learning has revolutionized video quality assessment by enabling automatic feature extraction and end-to-end learning. Bosse et al. [3] pioneered the application of convolutional neural networks (CNNs) for both full-reference and no-reference image quality assessment, demonstrating significant improvements over traditional handcrafted features.

The Neural Image Assessment (NIMA) framework [24] introduced aesthetic quality prediction using deep CNNs trained on large-scale datasets with human ratings. This work established the foundation for learning-based approaches that can capture complex perceptual quality attributes beyond technical distortions.

DOVER [30] represents a significant advancement in video quality assessment by introducing disentangled representation learning. The method computes video quality using fewer frames while maintaining assessment accuracy, addressing computational efficiency concerns in practical applications. DOVER employs separate pathways for aesthetic and technical quality assessment, allowing for more nuanced evaluation of different quality aspects.

Representative no-reference VQA baselines include VSFA[13], which couples deep frame features with GRU-based temporal modeling; FAST-VQA[29], which introduces fragment sampling and a transformer backbone for efficient end-to-end training; and STI-VQA [37], which explicitly models spatiotemporal interactions for user-generated videos. We use these works as canonical references when situating our design choices for temporal modeling and computational efficiency.

Recent work by Wu et al. [31] explored video quality assessment for user-generated content from both aesthetic and technical perspectives, highlighting the importance of considering multiple quality dimensions in modern video evaluation frameworks.

2.3. Multimodal Fusion and Attention Mechanisms

The integration of multiple modalities has become increasingly important in video understanding and assessment tasks. Attention mechanisms, introduced by Vaswani et al. [25], have proven effective in capturing long-range dependencies and enabling selective focus on relevant information across different modalities.

Hori et al. [9] pioneered attention-based multimodal fusion for video description, demonstrating how attention mechanisms can selectively attend to specific modalities (visual, motion, audio) and temporal segments. Their work established the foundation for cross-modal attention mechanisms that dynamically weight the importance of different information sources.

Hierarchical attention-based approaches [28] further advanced multimodal fusion by considering the intrinsic properties of different modalities and their hierarchical relationships. These methods recognize that different modalities contribute varying levels of information and require sophisticated fusion strategies to optimize their combination.

Recent advances in cross-modal conditional mechanisms [6] have focused on text-video retrieval applications, where textual semantic features serve as conditional projections to extract relevant visual features from video frames. These approaches demonstrate the importance of semantic alignment between textual descriptions and visual content.

2.4. AI-Generated Content Quality Assessment

The emergence of AI-generated content has created new challenges for quality assessment methodologies. Traditional metrics designed for natural images and videos often fail to capture the unique characteristics and artifacts present in AI-generated content. The semantic relationship between input prompts and generated visual outputs introduces additional complexity that requires specialized evaluation frameworks.

Recent competitions such as the NTIRE 2024 Quality Assessment of AI-Generated Content Challenge [15] have brought attention to this emerging research area. These challenges highlighted the need for comprehensive evaluation frameworks that can assess both technical quality and semantic alignment in AI-generated content.

The AIS 2024 Video Quality Assessment Challenge [4] focused on user-generated content but revealed similar challenges in developing robust quality metrics for diverse video content types. The competition emphasized the importance of computational efficiency and real-time assessment capabilities for practical applications.

Concurrently, CLIP-based blind AIGC IQA leverages vision–language alignment and multi-scale cues on images: CLIP-AGIQA proposes multimodal prompt learning guided by text–image consistency [7]; AMFF-Net fuses mixed-scale features to assess visual quality, authenticity, and prompt–image consistency [35]; and CLIP-DQA adapts CLIP with global–local views for domain-specific blind IQA [33]. Our work differs by targeting video AIGC and by learning quality-aware, text-conditioned fusion together with temporal modeling.

Current research in AI-generated content quality assessment faces several key challenges: (1) the need for multi-dimensional evaluation frameworks that consider temporal consistency, image fidelity, aesthetic appeal, and semantic alignment; (2) the lack of large-scale datasets with comprehensive human annotations across multiple quality dimensions; and (3) the computational complexity of processing high-resolution video content with sophisticated neural architectures.

2.5. Cross-Modal Understanding and Text-Video Alignment

Text-video alignment has become a critical component in evaluating AI-generated video content, where the semantic consistency between textual prompts and visual outputs determines the overall quality. Recent advances in vision-language models, particularly CLIP [20], have provided powerful tools for cross-modal understanding and representation learning.

The development of transformer-based architectures [5] has enabled more sophisticated modeling of spatiotemporal relationships in video content. These approaches leverage self-attention mechanisms to capture long-range dependencies and complex patterns that are essential for comprehensive quality assessment.

Recent work in multimodal alignment and fusion has focused on developing more effective strategies for combining information from different modalities. Attention-based fusion methods enable dynamic weighting of different information sources based on their relevance to the assessment task, leading to more accurate and interpretable quality predictions.

The integration of large language models and vision transformers has opened new possibilities for understanding the semantic relationship between textual descriptions and visual content. However, adapting these large-scale models for video quality assessment requires careful consideration of computational efficiency and task-specific fine-tuning strategies.

3. Method

Our approach presents two novel multi-modal architectures that leverage state-of-the-art video encoders combined with advanced text understanding capabilities. Both architectures are fundamentally designed to understand the relationship between textual prompts and the resulting video content, evaluating quality across four critical dimensions:

temporal consistency, image fidelity, aesthetic appeal, and text-video alignment.

3.1. DOVER++ with Quality-Aware Fusion

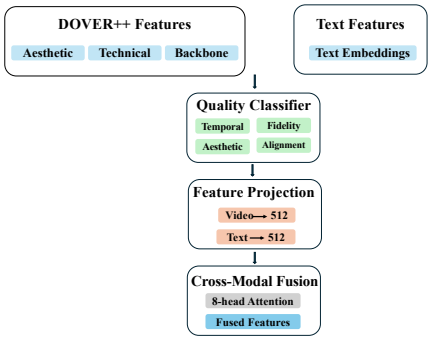


Figure 1. Quality-aware cross-modal fusion mechanism.

The DOVER++ architecture represents our first approach to multi-modal video quality assessment. We have significantly extended the original DOVER architecture [30] to incorporate textual understanding and cross-modal reasoning capabilities.

Our enhanced DOVER++ model consists of three primary components: a 3D ConvNeXt backbone [16] for video feature extraction, separate aesthetic and technical quality assessment heads, and a novel quality-aware fusion module for integrating text and video representations.

The backbone architecture employs a modified ConvNeXt structure adapted for 3D video processing. The feature extraction process can be mathematically formulated as:

$$\mathbf{F}_v = \text{ConvNeXt3D}(\mathbf{V}) \in \mathbb{R}^{B\times 768\times T'\times H'\times W'} \quad (1)$$

where $\mathbf{V} \in \mathbb{R}^{B\times 3\times T\times H\times W}$ represents the input video tensor with batch size B, temporal dimension $T = 64$, and spatial dimensions $H = W = 640$.

3.1.1. Quality-Aware Fusion Mechanism

The quality-aware fusion module represents our primary contribution to the DOVER++ architecture. Figure 1 illustrates the complete fusion process, which dynamically weights different quality aspects based on textual content. This module implements a sophisticated attention mechanism [25] that dynamically weights the importance of different quality aspects based on the textual prompt content.

The fusion process begins with a quality aspect classifier that analyzes the text embedding to determine which quality dimensions are most relevant:

$$\mathbf{w}_q = \text{Softmax}(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1\mathbf{t} + \mathbf{b}_1) + \mathbf{b}_2) \quad (2)$$

where $\mathbf{t} \in \mathbb{R}^{1024}$ is the text embedding from BGE-Large encoder [32], and $\mathbf{w}_q \in \mathbb{R}^4$ represents the attention weights for the four quality aspects.

As shown in Fig. 1, the cross-modal attention mechanism then aligns video and text features through a multi-head attention framework:

$$\mathbf{F}_{aligned} = \text{MultiHeadAttention}(\mathbf{Q}_t, \mathbf{K}_v, \mathbf{V}_v) \quad (3)$$

where $\mathbf{Q}_t$ is derived from projected text features, and $\mathbf{K}_v, \mathbf{V}_v$ are computed from projected video features.

3.2. V-JEPA2 with Discriminative Learning

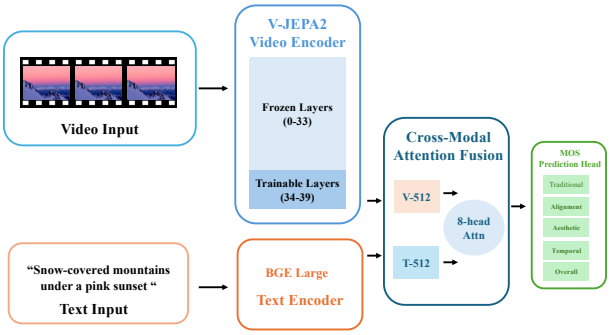


Figure 2. V-JEPA2 architecture with strategic layer freezing. The model employs 85% frozen layers and 15% trainable layers for efficient adaptation, followed by cross-modal attention fusion for video-text alignment. Input videos are processed at 384×384 resolution with 64 frames.

Our second architecture leverages V-JEPA2 (Video Joint Embedding Predictive Architecture) [2], originally developed for self-supervised video representation learning. We have adapted this architecture for supervised quality assessment through strategic modifications and training strategies.

The V-JEPA2 model processes video inputs at 384×384 resolution using a ViT-Giant architecture [5] with 64 frames per video. As illustrated in Fig. 2, our approach employs strategic layer freezing to achieve efficient adaptation of large-scale transformers. The feature extraction process can be represented as:

$$\mathbf{F}_{vjepa} = \text{ViT-G}(\text{Patch}(\mathbf{V})) \in \mathbb{R}^{B\times L\times 1408} \quad (4)$$

where $\text{Patch}(\mathbf{V})$ converts the video into spatiotemporal patches, L is the sequence length, and 1408 is the hidden dimension of the ViT-Giant model.

3.2.1. Strategic Layer Freezing

A critical innovation in our V-JEPA2 approach is the implementation of strategic layer freezing, inspired by recent

work in parameter-efficient fine-tuning [11]. As shown in Fig. 2, we freeze the bottom 85% of transformer layers (layers 0-33) while allowing the top 15% (layers 34-39) to adapt to the quality assessment task:

$$\frac{\partial \mathcal{L}}{\partial \theta_i} = \begin{cases} 0 & \text{if } i < 0.85 \times N_{layers} \\ \frac{\partial \mathcal{L}}{\partial \theta_i} & \text{otherwise} \end{cases} \tag{5}$$

where θ_i represents the parameters of the i-th layer and N_{layers} is the total number of transformer layers. This strategy reduces gradient computation by approximately 65% while maintaining model expressiveness.

3.2.2. Discriminative Learning Rate Strategy

The discriminative learning rate strategy [10] applies different learning rates to different model components based on their role and pre-training status:

$$\begin{aligned} \text{lr}_{text} &= 0.1 \times \text{lr}_{base} \\ \text{lr}_{video} &= 0.5 \times \text{lr}_{base} \\ \text{lr}_{head} &= 2.0 \times \text{lr}_{base} \end{aligned} \tag{6}$$

3.3. Hybrid Loss Function Design

Both architectures employ a sophisticated hybrid loss function that addresses multiple aspects of quality assessment. Our hybrid loss function combines three complementary components:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{smooth} + \beta \mathcal{L}_{rank} + \gamma \mathcal{L}_{scale} \tag{7}$$

The smooth L1 loss component [8] provides robust regression while being less sensitive to outliers:

$$\mathcal{L}_{smooth} = \frac{1}{N} \sum_{i=1}^{N} \text{SmoothL1}(\mathbf{y}_i, \hat{\mathbf{y}}_i) \tag{8}$$

The ranking loss ensures that the relative ordering of quality scores is preserved:

$$\mathcal{L}_{rank} = \frac{1}{N(N-1)} \sum_{i=1}^{N} \sum_{j \neq i} \max(0, \delta - (\hat{y}_i - \hat{y}_j) \cdot \text{sign}(y_i - y_j)) \tag{9}$$

where $\delta = 0.2$ is the margin parameter.

The scale-aware loss component addresses the importance of extreme quality values:

$$\mathcal{L}_{scale} = \frac{1}{N} \sum_{i=1}^{N} w_i \cdot (\mathbf{y}_i - \hat{\mathbf{y}}_i)^2 \tag{10}$$

where $w_i = 1.5$ if $y_i < 2.5$ or $y_i > 4.0$, and $w_i = 1.0$ otherwise.

For the V-JEPA2 architecture, we implement adaptive loss weighting [12] that dynamically adjusts the loss coefficients during training:

$$\alpha_{t+1} = \alpha_t - \eta \cdot \text{sign}(\mathcal{L}_{smooth,t} - \mathcal{L}_{rank,t}) \tag{11}$$

where $\eta = 0.1$ is the adaptation rate and t denotes the training iteration.

4. Experiments

4.1. Dataset and Evaluation Metrics

We evaluate our approaches on the TaobaoVD-GC dataset provided for the VQualA 2025 challenge, which contains 4,000 training videos with corresponding quality annotations across four dimensions: temporal consistency, image fidelity, aesthetic appeal, and text-video alignment. Additionally, a test set of 500 videos was provided for final evaluation.

4.1.1. Evaluation Methodology

The VQualA 2025 challenge employs a comprehensive multi-dimensional quality assessment framework that evaluates AI-generated videos across five key metrics: four sub-dimensional Mean Opinion Scores (MOS) and one overall MOS. Each video in the dataset is annotated with human-rated quality scores for the following dimensions:

Temporal Consistency MOS: Evaluates the smoothness of motion and coherence of scene transitions across video frames. This metric captures artifacts such as temporal flickering, inconsistent object motion, and abrupt scene changes that disrupt visual continuity.

Image Fidelity MOS: Assesses the photorealistic quality and technical fidelity of individual video frames. This includes evaluation of visual artifacts, resolution quality, color accuracy, and overall image clarity.

Aesthetic Appeal MOS: Measures the artistic and visual attractiveness of the video content. This dimension considers composition, lighting, color harmony, and overall visual appeal that contributes to viewer engagement.

Text-Video Alignment MOS: Evaluates how well the generated video content corresponds to the input textual description. This critical metric assesses semantic consistency between the prompt and visual output, including object presence, scene setting, and action correspondence.

Overall MOS: Provides a holistic quality assessment that considers all aspects of video quality. This metric represents the general human perception of video quality across all dimensions.

The evaluation methodology follows the challenge protocols, utilizing Spearman's Rank Correlation Coefficient (SROCC) and Pearson Linear Correlation Coefficient (PLCC) as primary correlation metrics. For each of the five MOS dimensions, both SROCC and PLCC are computed between predicted scores and ground truth human annotations. The final challenge score aggregates performance across all dimensions:

$$\text{Final Score} = \frac{\text{SROCC}_{overall} + \text{PLCC}_{overall}}{2} \tag{12}$$

where $\text{SROCC}_{overall}$ and $\text{PLCC}_{overall}$ represent the correlation coefficients computed on the overall MOS dimension. This evaluation framework ensures that models must demonstrate competence across all quality aspects while achieving strong correlation with human perceptual judgments on the overall quality assessment.

4.2. Implementation Details

4.2.1. Data Preparation

For the DOVER++ approach, we employed a 90:10 train-validation split from the 4,000 training videos, resulting in 3,600 training samples and 400 validation samples. For the V-JEPA2 approach, we used an 80:20 split, yielding 3,200 training samples and 800 validation samples. This difference in split ratios was motivated by the higher model complexity of V-JEPA2, which benefits from more validation data for robust hyperparameter tuning. Final evaluation is performed on the official test set of 500 videos.

The dataset preparation involves uniform temporal sampling of 64 frames from each video, ensuring consistent temporal coverage regardless of the original video length. Spatial preprocessing includes resizing to target resolutions (640×640 for DOVER++, 384×384 for V-JEPA2) with careful preservation of aspect ratios to prevent distortion artifacts.

4.2.2. Training Configuration

For the DOVER++ architecture, we use a batch size of 4 with gradient accumulation over 8 steps, resulting in an effective batch size of 32. The learning rate is set to 1×10^{-4} with cosine annealing scheduling over 5 epochs. Training is performed on NVIDIA H200 GPUs with 143.7GB memory.

The V-JEPA2 architecture requires more sophisticated training strategies due to its larger parameter count (1.37B total parameters). We employ a batch size of 6 with gradient accumulation over 32 steps, achieving an effective batch size of 192. The discriminative learning rate strategy uses a base learning rate of 2×10^{-4} with component-specific multipliers as described in Section 3.

Both architectures utilize AdamW optimization [17] with weight decay of 1×10^{-2} for regularization. Gradient clipping with a maximum norm of 1.0 prevents gradient explosion. We employ mixed-precision training [18] using FP16 for forward passes while maintaining FP32 precision for gradient computations.

4.3. Computational Complexity

Table 1 presents the technical specifications for both approaches. The DOVER++ approach achieves excellent performance with moderate computational requirements, making it suitable for real-time applications. The V-JEPA2 approach, while computationally more intensive, delivers superior quality assessment accuracy through its sophisticated transformer-based processing.

4.4. Baseline and Alternative Methods

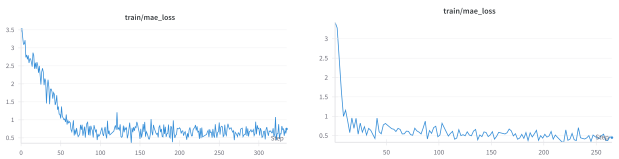
During the development process, we explored several alternative approaches to validate our architectural choices. We investigated traditional CNN-based video encoders (ResNet3D, I3D), different text encoders (CLIP, RoBERTa), and various fusion strategies (simple concatenation, element-wise multiplication, attention pooling). These baseline experiments confirmed that our proposed quality-aware fusion mechanism and discriminative learning strategies provide clear advantages for the multi-modal video quality assessment task.

We also investigated ensemble methods combining multiple models, but found that the computational overhead outweighed the marginal performance gains. The focus on individual model optimization through advanced training strategies proved more effective for this challenge.

5. Results and Analysis

5.1. Quantitative Results

Table 2 presents the performance of our two architectures on both validation and test sets of the VQualA 2025 challenge. The V-JEPA2 approach with discriminative learning demonstrates superior performance compared to the DOVER++ approach with quality-aware fusion, achieving notable improvements in both validation and test scenarios.



(a) DOVER++ training MAE loss (b) V-JEPA2 training MAE loss

Figure 3. Training loss curves demonstrating convergence behavior for both architectures.

Figure 3 illustrates the training convergence behavior for both architectures, providing crucial insights into the learning dynamics. Both models demonstrate stable learning with consistent loss reduction throughout training, validating the effectiveness of our architectural designs and training strategies. The DOVER++ architecture (Figure 3a) shows smooth convergence with the MAE loss decreasing steadily from approximately 3.5 to 0.5 over 250 training steps. The V-JEPA2 model (Figure 3b) exhibits robust learning dynamics with loss reduction from 3.5 to approximately 0.7 over 300 steps.

The results demonstrate the effectiveness of our multi-modal approaches for AI-generated video quality assessment. The V-JEPA2 architecture's superior performance

Method	Resolution	Training Time	GFLOPs	Params (M)	Runtime (ms)
DOVER++	640×640×64	7.5h	850	370	800
V-JEPA2	384×384×64	15h	1200	1370	1000

Table 1. Technical specifications for both approaches. Training performed on NVIDIA H200 GPUs.

Method	Validation Score	Test Score
DOVER++	0.52	0.39
V-JEPA2	0.63	0.57

Table 2. Final scores on validation and test sets for both architectures.

can be attributed to its sophisticated transformer-based processing and strategic layer freezing, which enables effective adaptation of large-scale pre-trained models to the quality assessment task.

5.2. Ablation Studies

We conducted comprehensive ablation studies to understand the contribution of different components in our architectures. Table 3 presents the final score values for various architectural configurations, evaluated on our held-out validation data.

The ablation studies reveal the importance of each architectural component in achieving optimal performance. For the DOVER++ architecture, the quality-aware cross-modal fusion mechanism provides the most significant improvement (+0.14), demonstrating the critical importance of effectively combining video and text representations. The hybrid loss function contributes an additional +0.02 improvement, while discriminative learning rates provide a final +0.02 boost, resulting in a total improvement of 0.18 over the baseline.

For the V-JEPA2 architecture, strategic layer freezing yields the largest performance gain (+0.15), confirming the effectiveness of our parameter-efficient fine-tuning approach for large-scale transformers. The hybrid loss function adds +0.02 improvement, and the complete integration of all components provides an additional +0.01 refinement.

Both architectures show consistent positive contributions from each component, with the cross-modal fusion and layer freezing strategies providing the most substantial improvements. The hybrid loss function proves beneficial for both approaches, while discriminative learning rates offer additional fine-tuning benefits for the DOVER++ model.

5.3. Qualitative Analysis

Our architectures demonstrate distinct strengths in handling different aspects of video quality assessment. The quality-aware fusion mechanism in DOVER++ enables dynamic attention allocation based on textual content, allowing the model to focus on relevant quality aspects depending on the input prompt characteristics.

The V-JEPA2 architecture excels at capturing long-range temporal dependencies and complex spatiotemporal patterns crucial for comprehensive quality assessment.

5.4. Computational Efficiency

The computational analysis reveals important trade-offs between performance and efficiency across our two approaches. Table 1 in Section 4 provides detailed computational specifications for both architectures.

The DOVER++ approach offers an excellent balance between performance and computational efficiency, requiring 850 GFLOPs and 370M parameters while achieving a validation score of 0.52. This makes it particularly suitable for applications where computational resources are limited or real-time processing is required.

The V-JEPA2 approach, while requiring significantly more computational resources (1200 GFLOPs and 1370M parameters), delivers superior quality assessment accuracy with a validation score of 0.63. The reduced parameter count requiring gradient updates improves training efficiency while maintaining performance.

5.5. Error Analysis and Limitations

Our analysis reveals several patterns in model performance across different video characteristics. Both architectures show consistent performance across the four quality dimensions evaluated in the challenge, with particular strengths in temporal consistency and text-video alignment assessment.

The models demonstrate robust performance across diverse video content types, effectively handling variations in visual complexity, motion patterns, and semantic content. However, certain edge cases and extreme quality scenarios remain challenging, suggesting opportunities for future improvements in model robustness and generalization capabilities.

6. Conclusion

This work presents QualiVision, a novel multi-modal approach for video quality assessment that addresses the unique challenges of AI-generated content evaluation. Our

Component Configuration	DOVER++ Score	V-JEPA2 Score
Baseline (Multi-modal adapted architecture)	0.34	0.45
+ Quality-Aware Cross-Modal Fusion / + Strategic Layer Freezing	0.48	0.60
+ Hybrid Loss Function	0.50	0.62
+ Discriminative Learning Rates	0.52	0.63
Full Model (All Components)	**0.52**	**0.63**

Table 3. Ablation study results showing the contribution of different components to the final validation scores.

contribution introduces two complementary architectures that leverage state-of-the-art video encoders combined with advanced text understanding capabilities.

The quality-aware cross-modal fusion mechanism in our DOVER++ architecture enables dynamic attention allocation based on textual content, while the discriminative learning strategy in V-JEPA2 optimizes training efficiency for large-scale models. The hybrid loss function addresses multiple aspects of quality perception, ensuring that model predictions align with human judgments across different quality dimensions.

Our experimental results on the VQualA 2025 challenge demonstrate significant improvements over traditional video quality metrics, with the V-JEPA2 architecture achieving a test score of 0.57 and the DOVER++ approach obtaining 0.39. These results represent substantial advances in multi-modal video quality assessment and establish new benchmarks for AI-generated content evaluation.

The architectural innovations presented in this work demonstrate the importance of cross-modal understanding in video quality assessment, particularly for AI-generated content where text-video alignment is crucial. The strategic layer freezing and discriminative learning rate strategies provide practical templates for efficient fine-tuning of large-scale pre-trained models.

For reproducibility, we release our code and models at: https://github.com/RITIK-12/QualiVision.

6.1. Limitations and Future Work

While our approaches show promising results, several limitations merit consideration. The models occasionally exhibit sensitivity to textual phrases that may not directly relate to visual quality, suggesting opportunities for developing more robust text understanding mechanisms. Additionally, performance on videos with extreme quality scores remains challenging, indicating potential for improved loss function design. Finally, we do not include standardized comparisons with widely used no-reference VQA baselines under an identical protocol, which we leave for future work.

Future work could explore several directions: (1) developing more sophisticated attention mechanisms that can better distinguish between quality-relevant and quality-irrelevant textual content, (2) investigating ensemble methods that combine complementary architectures while maintaining computational efficiency, (3) extending the approach to handle longer video sequences and higher resolutions, and (4) exploring applications to other types of generated content beyond videos.

The integration of emerging foundation models and the exploration of self-supervised learning strategies for quality assessment represent promising avenues for advancing this field. Additionally, the development of interpretability tools that can provide insights into model decision-making processes would enhance the practical applicability of these approaches.

6.2. Broader Impact

Our work contributes to the broader goal of developing reliable quality assessment tools for AI-generated content, which is increasingly important as generative models become more prevalent in various applications. The ability to automatically assess the quality of generated videos has implications for content creation, media production, and quality control in AI-assisted workflows.

The multi-modal nature of our approach provides a foundation for developing more comprehensive evaluation frameworks that consider both technical and semantic aspects of content quality. This work represents a step toward bridging the gap between computational metrics and human perceptual quality judgments in the era of AI-generated media.

References

[1] ArtSmart. Ai video generator market statistics for 2025, 2025. https://artsmart.ai/blog/ai-video-generator-statistics/. 1

[2] Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nico-

las Ballas. V-jepa 2: Self-supervised video models enable understanding, prediction and planning, 2025. 2, 4

[3] Sebastian Bosse, Dominique Maniry, Klaus-Robert Müller, Thomas Wiegand, and Wojciech Samek. Deep neural networks for no-reference and full-reference image quality assessment. *IEEE Trans. Image Process.*, 27(1):206–219, 2017. 2

[4] Marcos V. Conde, Saman Zadtootaghaj, Nabajeet Barman, Radu Timofte, Chenlong He, Qi Zheng, Ruoxi Zhu, Zhengzhong Tu, Haiqiang Wang, Xiangguang Chen, Wenhui Meng, Xiang Pan, Huiying Shi, Han Zhu, Xiaozhong Xu, Lei Sun, Zhenzhong Chen, Shan Liu, Zicheng Zhang, Haoning Wu, Yingjie Zhou, Chunyi Li, Xiaohong Liu, Weisi Lin, Guangtao Zhai, Wei Sun, Yuqin Cao, Yanwei Jiang, Jun Jia, Zhichao Zhang, Zijian Chen, Weixia Zhang, Xiongkuo Min, Steve Göring, Zihao Qi, and Chen Feng. Ais 2024 challenge on video quality assessment of user-generated content: Methods and results, 2024. 1, 3

[5] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *Int. Conf. Learn. Represent.*, 2021. 3, 4

[6] Wanru Du, Xiaochuan Jing, Quan Zhu, Xiaoyin Wang, and Xuan Liu. A cross-modal conditional mechanism based on attention for text-video retrieval. *Mathematical Biosciences and Engineering*, 20(11):20073–20092, 2023. 3

[7] Jun Fu, Wei Zhou, Qiuping Jiang, Hantao Liu, and Guangtao Zhai. Vision-language consistency guided multi-modal prompt learning for blind ai generated image quality assessment. *arXiv preprint arXiv:2406.16641*, 2024. 3

[8] Ross Girshick. Fast r-cnn. In *Int. Conf. Comput. Vis.*, pages 1440–1448, 2015. 5

[9] Chiori Hori, Takaaki Hori, Teng-Yok Lee, Kazuhiro Sumi, John R. Hershey, and Tim K. Marks. Attention-based multimodal fusion for video description, 2017. arXiv preprint arXiv:1701.03126. 3

[10] Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. In *Annu. Meet. Assoc. Comput. Linguist.*, pages 328–339, 2018. 5

[11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *Int. Conf. Learn. Represent.*, 2022. 5

[12] Alex Kendall, Yarin Gal, and Roberto Cipolla. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 7482–7491, 2018. 5

[13] Dingquan Li, Tingting Jiang, and Ming Jiang. Quality assessment of in-the-wild videos. In *Proceedings of the 27th ACM International Conference on Multimedia (MM)*, pages 2351–2359, 2019. 2

[14] Zhi Li, Anne Aaron, Ioannis Katsavounidis, Anush Moorthy, and Megha Manohara. Toward a practical perceptual video quality metric. Netflix Technology Blog, 2016. Netflix Technology Blog post introducing VMAF. 2

[15] Xiaohong Liu, Xiongkuo Min, Guangtao Zhai, Chunyi Li, Tengchuan Kou, Wei Sun, Haoning Wu, Yixuan Gao, Yuqin Cao, Zicheng Zhang, et al. Ntire 2024 quality assessment of ai-generated content challenge. In *IEEE Conf. Comput. Vis. Pattern Recog. Worksh.*, 2024. 1, 3

[16] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 11976–11986, 2022. 4

[17] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *Int. Conf. Learn. Represent.*, 2019. 6

[18] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, et al. Mixed precision training. In *Int. Conf. Learn. Represent.*, 2018. 6

[19] Xiongkuo Min, Guangtao Zhai, Ke Gu, and Xiaokang Yang. Perceptual video quality assessment: a survey. *Science China Information Sciences*, 2024. 2

[20] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Int. Conf. Machine Learn.*, pages 8748–8763, 2021. 3

[21] Grand View Research. Ai video generator market size and share report, 2030, 2024. https://www.grandviewresearch.com/industry-analysis/ai-video-generator-market-report. 1

[22] Kalpana Seshadrinathan, Rajiv Soundararajan, Alan C. Bovik, and Lawrence K. Cormack. Study of subjective and objective quality assessment of video. *IEEE Trans. Image Process.*, 19(6):1427–1441, 2010. 1, 2

[23] Hamid R. Sheikh and Alan C. Bovik. Image information and visual quality. *IEEE Trans. Image Process.*, 15(2):430–444, 2006. 2

[24] Hossein Talebi and Peyman Milanfar. Nima: Neural image assessment. *IEEE Trans. Image Process.*, 27(8):3998–4011, 2018. 2

[25] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Adv. Neural Inform. Process. Syst.*, pages 5998–6008, 2017. 3, 4

[26] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. Image Process.*, 13(4): 600–612, 2004. 2

[27] Stefan Winkler. Perceptual video quality metrics — a review. In *Digital Video Image Quality and Perceptual Coding*. CRC Press, 2006. 2

[28] Chunlei Wu, Yiwei Wei, Xiaoliang Chu, Sun Weichen, Fei Su, and Leiquan Wang. Hierarchical attention-based multimodal fusion for video captioning. *Neurocomputing*, 315: 354–364, 2018. 3

[29] Haoning Wu, Chaofeng Chen, Jingwen Hou, Liang Liao, Annan Wang, Wenxiu Sun, Qiong Yan, and Weisi Lin. Fast-vqa: Efficient end-to-end video quality assessment with fragment sampling. In *Computer Vision – ECCV 2022*, pages 538–554. Springer, 2022. 2

[30] Haoning Wu, Erli Zhang, Liang Liao, Chaofeng Chen, Jingwen Hou, Annan Wang, Wenxiu Sun, Qiong Yan, and Weisi Lin. Exploring video quality assessment on user generated contents from aesthetic and technical perspectives. In *Int. Conf. Comput. Vis.*, pages 20087–20097, 2023. 2, 4

[31] Haoning Wu, Zicheng Zhang, Weixia Zhang, Chaofeng Chen, Liang Liao, Chunyi Li, Yixuan Gao, Annan Wang, Erli Zhang, Wenxiu Sun, et al. Exploring video quality assessment on user generated contents from aesthetic and technical perspectives, 2023. 2

[32] Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. C-pack: Packaged resources to advance general chinese embedding, 2023. 4

[33] Yirui Zeng, Jun Fu, Hadi Amirpour, Huasheng Wang, Guanghui Yue, Hantao Liu, Ying Chen, and Wei Zhou. Clip-dqa: Blindly evaluating dehazed images from global and local perspectives using clip. *arXiv preprint arXiv:2502.01707*, 2025. 3

[34] Qi Zheng, Yibo Fan, Leilei Huang, Tianyu Zhu, Jiaming Liu, Zhijian Hao, Shuo Xing, Chia-Ju Chen, Xiongkuo Min, Alan C. Bovik, and Zhengzhong Tu. Video quality assessment: A comprehensive survey, 2024. 2

[35] Tianwei Zhou, Songbai Tan, Wei Zhou, Yu Luo, Yuan-Gen Wang, and Guanghui Yue. Adaptive mixed-scale feature fusion network for blind ai-generated image quality assessment. *arXiv preprint arXiv:2404.15163*, 2024. 3

[36] Wei Zhou, Hadi Amirpour, Christian Timmerer, Guangtao Zhai, Patrick Le Callet, and Alan C. Bovik. Perceptual visual quality assessment: Principles, methods, and future directions. *arXiv preprint arXiv:2503.00625*, 2025. 2

[37] Hanwei Zhu, Baoliang Chen, Lingyu Zhu, and Shiqi Wang. Learning spatiotemporal interactions for user-generated video quality assessment. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(3):1031–1042, 2023. 2